

Linear Algebra – The Backbone of AI/ML

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TOPIC :

PROBLEM SOLVING ON LA

Sanjay Kumar Sen

Assistant Professor at Quad AI



Assessment for Theory Courses

Quiz - 10 M

Internal - 50 M

Mid-Term Assessment - 20 M
(10 M x 2)

Assignment - 15 M (7.5 M x 2)

Attendance - 5

End Term Assessment - 50 M

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External - 50 M.

Symmetric Matrices ($A^T = A$)

Skew-Symmetric Matrices ($A^T = -A$)

Orthogonal Matrices ($A^T = A^{-1}$ or $AA^T = I$)

Prob-1

Given The matrices $J = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 6 \end{bmatrix}$

and $K = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$, The product $K^T J K$ is —

Solⁿ:-

$$\begin{aligned} K^T J K &= \begin{bmatrix} 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} 6 & 8 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \\ &= 6 + 16 + 1 = 23, \end{aligned}$$

Prob-2

For $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$, The determinant of $A^T A^{-1}$ is -

(a) $\sec^2 x$

(b) $\cos 4x$

(c) 1

(d) 0

Prob-3 The inverse of the 2×2 matrix

$$\begin{bmatrix} 1 & 2 \\ 5 & 7 \end{bmatrix} \text{ is -}$$

- ✓ (a) $\frac{1}{3} \begin{bmatrix} -7 & 2 \\ 5 & -1 \end{bmatrix}$ (b) $\frac{1}{3} \begin{bmatrix} 7 & 2 \\ 5 & 1 \end{bmatrix}$
(c) $\frac{1}{3} \begin{bmatrix} 7 & -2 \\ 5 & 1 \end{bmatrix}$ (d) $\frac{1}{3} \begin{bmatrix} -7 & -2 \\ -5 & -1 \end{bmatrix}$

Solⁿ:

Inverse of $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 5 & 7 \end{bmatrix}^{-1} = \frac{1}{(7-10)} \begin{bmatrix} 7 & -2 \\ -5 & 1 \end{bmatrix}$$

$$= \frac{1}{-3} \begin{bmatrix} 7 & -2 \\ -5 & 1 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} -7 & 2 \\ 5 & -1 \end{bmatrix}$$

Prob-4

$$\text{Let, } A = \begin{bmatrix} 2 & -0.1 \\ 0 & 3 \end{bmatrix} \text{ and } A^{-1} = \begin{bmatrix} \frac{1}{2} & a \\ 0 & b \end{bmatrix}$$

Then, $(a+b) = ?$

- ✓ (a) $\frac{7}{20}$ (b) $\frac{3}{20}$ (c) $\frac{19}{60}$ (d) $\frac{11}{20}$

Solⁿ:-

$$[AA^{-1}] = I$$

$$\Rightarrow \begin{bmatrix} 2 & -0.1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & a \\ 0 & b \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 2a - 0.1b \\ 0 & 3b \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$2a - 0.1b = 0$$

$$\Rightarrow a = 0.1b/2$$

$$3b = 1 \Rightarrow \boxed{b = \frac{1}{3}}$$

Now, Substitute 'b' in equation (i)

We get, $a = \frac{1}{60}$

$$\text{So, } a+b = \frac{1}{60} + \frac{1}{3} = \frac{1+20}{60} = \frac{21}{60} = \frac{7}{20}$$

Prob-5

The eigenvectors of the matrix $\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$ are written in the form $\begin{bmatrix} a \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ b \end{bmatrix}$. What is $a+b$?

- (a) 0 (b) $1/2$
(c) 1 (d) 2

Solⁿ:- $\begin{vmatrix} (1-\lambda) & 2 \\ 0 & (2-\lambda) \end{vmatrix} = 0$
 $(1-\lambda)(2-\lambda) = 0$
 $\lambda = 1, 2$

Now, since the eigenvalue problem is,

$$[A - \lambda I]x = 0$$

$$\begin{vmatrix} 1-\lambda & 2 \\ 0 & 2-\lambda \end{vmatrix} x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Putting the value of $\lambda = 1$ and $x = x_1$
 $= \begin{bmatrix} a \\ 1 \end{bmatrix}$

$$\begin{bmatrix} 0 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ a \end{bmatrix} = 0$$

$$a = 0$$

Putting the value of $\lambda = 2$ and $x = x_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$\begin{bmatrix} -1 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ b \end{bmatrix} = 0$$

$$-1 + 2b = 0$$

$$b = \frac{1}{2}$$

From, $a + b = 0 + \frac{1}{2} = \frac{1}{2}$

Prob-6

For the matrix $\begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$ the eigen-value corresponding to the eigenvector $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is -

(A) 2
(C) 6

(B) 4
(D) 8

Sol:-

$$M = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}, M - \lambda I = \begin{bmatrix} 4 - \lambda & 2 \\ 2 & 4 - \lambda \end{bmatrix}$$

Given eigen vector $\begin{bmatrix} 101 \\ 101 \end{bmatrix}$ ✓

$$[M - \lambda I]x = 0$$

$$\begin{bmatrix} 4-\lambda & 2 \\ 2 & 4-\lambda \end{bmatrix} \begin{bmatrix} 101 \\ 101 \end{bmatrix} = 0$$

$$(4-\lambda)(101) + 2 \times 101 = 0$$
$$\lambda = 6$$

Prob-7

What are the eigen values of the following 2×2 matrix? $\begin{bmatrix} 2 & -1 \\ -4 & 5 \end{bmatrix}$

(a) -1 and 1

✓ (b) 1 and 6 -

(c) 2 and 5

(d) 4 and -1 -

Sol:-

$$A = \begin{bmatrix} 2 & -1 \\ -4 & 5 \end{bmatrix}$$

The characteristic eq of this matrix is given by,

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & -1 \\ -4 & 5-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)(5-\lambda)-4=0$$

$$\lambda^2-7\lambda+6=0$$

$$\lambda=1,6$$

$\therefore$ The eigen values of A are -
1 and 6,

Prob-8 The sum of Eigen values of matrix, [M] is

Where $[M] = \begin{bmatrix} 215 & 650 & 795 \\ 655 & 150 & 835 \\ 485 & 355 & 550 \end{bmatrix}$

☒ (A) 915

☐ (B) 1355

☐ (C) 1640

☐ (D) 2180

Sol? - Sum of eigen values
= trace of Matrix
= $215 + 150 + 550$
= 915 Am

Prob-9

$$A = \begin{bmatrix} -3 & 2 \\ -1 & 0 \end{bmatrix}$$

A satisfies the relation (Apply C-H Theorem) -

☐ (a) $A + 3I + 2A^{-1} = 0$

$$(b) A^2 + 2A + 2I = 0$$

$$(c) (A+I)(A+2I) = I$$

$$(d) \exp(A) = 0$$

Solⁿ:-

$$A = \begin{bmatrix} -3 & 2 \\ -1 & 0 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} -3-\lambda & 2 \\ -1 & 0-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (-3-\lambda)(-\lambda) + 2 = 0$$

$$\lambda^2 + 3\lambda + 2 = 0$$

A will satisfy this equation according to Cayley-Hamilton Theorem-

$$\text{i.e. } A^2 + 3A + 2I = 0$$

Multiplying by A^{-1} on both sides, we get

$$A^{-1}A^2 + 3A^{-1}A + 2A^{-1}I = 0$$

$$\boxed{A + 3I + 2A^{-1} = 0}$$

Prob - 10

A^9 equals to -

(a) $511A + 5101$

(b) $309A + 1041$

(c) $154A + 1551$

(d) $\exp(9A)$ -

Connected with above Question
— Homework —

← Happy Learning →